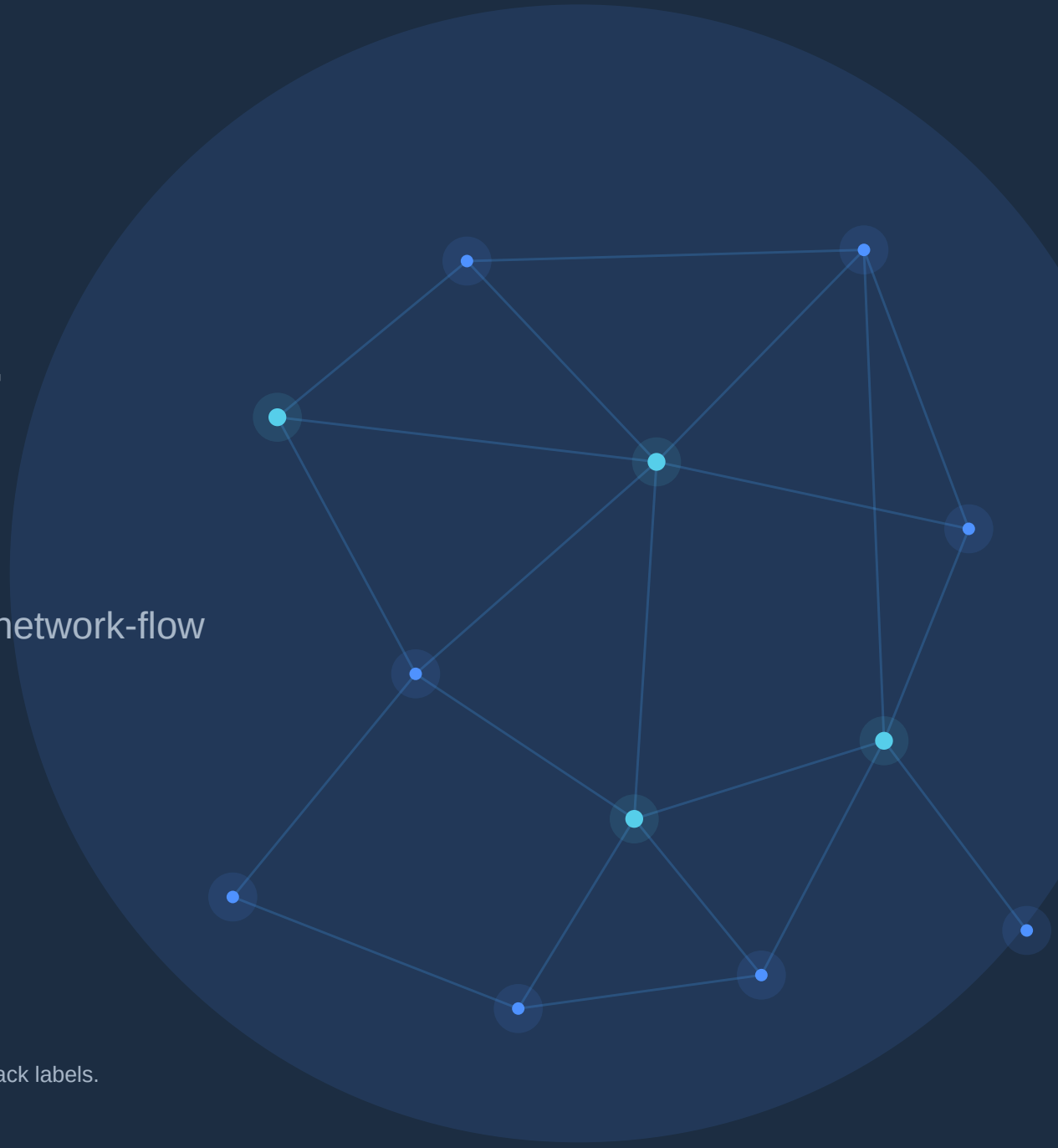
EXPERT BRIEFING

Explainable Cyber Threat Defense Workbench

Comparing detection signals and explainable models for network-flow attack detection.



UNSW-NB15 | HELD-OUT EVALUATION | EXPLAINABLE ML

A focused research workflow for understanding model performance, alert volume, and missed attack labels.

A clear question. A focused comparison.

Can readable detection signals help models rank attack-labeled flows while making review tradeoffs easier to understand?

73,791
primary-test flows

51.9%
attack-labeled

3 x 3
signals and classifiers

The three detection tracks are evaluated on their own and also supply signals to each classifier. XGBoost was selected using validation average precision before the primary test comparison.

DATA SPLIT Published training and test files; 8,541 exact feature-overlap test rows excluded from the primary test.

XGBoost led the held-out model comparison.

Model selection used validation average precision; all reported metrics below are from the primary test.

LOGISTIC REGRESSION

Average precision	ROC AUC	BRIER
0.9696	0.9654	0.1088

XGBOOST

Average precision	ROC AUC	BRIER
0.9825	0.9796	0.0929

EXPLAINABLE BOOSTING MACHINE

Average precision	ROC AUC	BRIER
0.9746	0.9710	0.1049

SELECTED MODEL

XGBoost had the strongest primary-test AP and ROC AUC, and the lowest Brier score.

At the same review volume, XGBoost found more attack labels.

Rules union and XGBoost were compared at exactly 27,973 alerts on the primary test.

RULES UNION

27,973

ALERTS

22,578

attack-labeled flows

5,395

normal-labeled flows

PRECISION 80.7%

RECALL 59.0%

XGBOOST

27,973

ALERTS

27,899

attack-labeled flows

74

normal-labeled flows

PRECISION 99.7%

RECALL 72.9%

+5,321

additional attack-labeled flows in the model queue

The model queue also contains 5,321 fewer normal-labeled flows at this same alert count.

Benchmark counts at a matched queue size; the project compares flow labels, not incident counts.

Each detection track surfaces a different pattern.

Standalone track alerts at the validation-set 90th-percentile cutoff for scored tracks.

READABLE RULES

27,973

PRIMARY-TEST ALERTS

22,578	80.7%
attack-labeled	precision

RECALL

59.0%

Transparent fixed conditions

ISOLATION FOREST

5,889

PRIMARY-TEST ALERTS

4,039	68.6%
attack-labeled	precision

RECALL

10.6%

Novelty score fit on normal-labeled rows

CONNECTION CONCENTRATION

7,157

PRIMARY-TEST ALERTS

7,157	100.0%
attack-labeled	precision

RECALL

18.7%

Uses supplied recent-connection counts

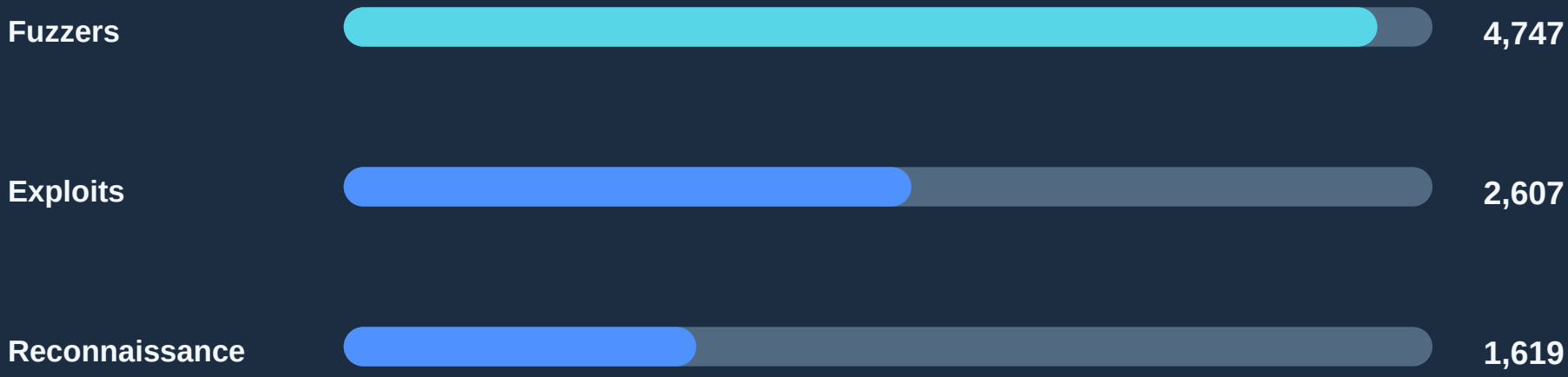
BENCHMARK OBSERVATION

Connection concentration had 100% measured precision in this primary test.

This result is specific to the evaluated UNSW-NB15 rows.

The missed-attack review points to clear follow-up areas.

At the rule-matched model alert budget, XGBoost missed 10,364 attack-labeled flows.



MODEL EXPLANATIONS

Global view

Feature importance compares which inputs shape each model.

Local view

A SHAP example shows feature contributions to one missed-flow score.

Analyst value

Misses and score drivers become concrete review targets.

Attack categories and counts are benchmark labels; explanations describe model scores.

A compact, explainable workflow for network-flow research.

The project connects detection signals, model ranking, analyst review volume, and missed-category analysis.

01

Compare models

Three classifier families use shared flow inputs and detection signals.

02

Measure workload

A matched alert count makes the rule-versus-model comparison interpretable.

03

Inspect misses

Attack-category counts and SHAP contributions show where review can continue.

STUDY FRAME

UNSW-NB15

Published cyber-range network-flow dataset. Primary test: 73,791 rows after removing exact train/test feature overlap.

The published files have no timestamps. Fit and validation use grouped random splitting; results describe this benchmark partition.

SOURCE

Moustafa and Slay, UNSW-NB15: a comprehensive data set for network intrusion detection systems, MilCIS 2015. DOI: 10.1109/MilCIS.2015.7348942. Dataset: research.unsw.edu.au/projects/unsw-nb15-dataset